

Predicting Diabetes Readmissions in U.S Hospitals

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Introduction

Hospital readmissions among patients with diabetes pose a substantial challenge to the U.S. healthcare system, contributing to elevated costs, increased resource utilization, and heightened risks of adverse clinical outcomes (Gupta et al., 2018). As one of the most prevalent chronic conditions in the United States, diabetes requires continuous and coordinated care. Patients frequently experience complications that require hospital visits, making the identification of risk factors a critical public health priority.

Advances in data analytics and machine learning have created opportunities to better understand the drivers of readmission and to support evidence-based decision-making in clinical settings (Sarker, 2021). Predictive modeling enables the systematic analysis of large, complex electronic health record (EHR) datasets to identify patterns, quantify risk, and enhance early intervention strategies. When applied effectively, such models can inform population-level management approaches, guide personalized clinical care and reduce preventable readmissions.

This report examines the development and evaluation of predictive models aimed at estimating the likelihood of hospital readmission among diabetic patients within U.S. hospitals. Utilizing the `diabetic_data.csv` dataset, which contains detailed demographic, diagnostic, procedural, and pharmaceutical information, we investigate key predictors of readmission and assess the comparative performance of several machine learning algorithms. The objective of this study is to generate empirically grounded insights that may support improved care coordination, facilitate targeted intervention strategies, and ultimately contribute to reduced readmission rates for diabetic populations.

Background & Related Work

Diabetes continues to be one of the most prevalent and costly chronic diseases in the United States, driving substantial healthcare utilization and placing considerable strain on hospital systems (Gupta et al., 2018). A major contributor to this burden is the high rate of hospital readmissions among diabetic patients, often linked to complications such as cardiovascular disease, renal dysfunction, poor glycemic control, and polypharmacy. Thirty-day readmission rates are widely used as indicators of care quality and are monitored by organizations such as the Centers for Medicare & Medicaid Services (CMS), which ties hospital reimbursement to performance on readmission metrics. Understanding the clinical, demographic, and treatment-related factors associated with diabetes readmission is therefore essential for improving patient outcomes and directing resources toward at-risk populations.

Recent research has increasingly leveraged electronic health record (EHR) data to identify predictors of hospital readmission and develop risk-stratification tools. Studies using traditional statistical models have highlighted factors such as age, comorbid diagnoses, admission type, length of stay, and

medication patterns as important determinants of readmission (Dinh et al., 2019). However, the complexity of interactions among clinical variables has motivated a shift toward machine-learning-based approaches, which can better capture nonlinear relationships and higher-dimensional patterns in patient data (Sarker, 2021).

Prior studies using similar EHR-based datasets have demonstrated that variables such as number of prior inpatient visits, changes in diabetes medication, and the intensity of laboratory testing can be strong predictors of readmission risk (Dinh et al., 2019). Machine learning models such as random forests, gradient boosting machines, and logistic regression with interaction terms have shown promising predictive performance, though challenges remain in addressing class imbalance and ensuring interpretability. Moreover, recent literature emphasizes the importance of integrating feature engineering such as the interaction terms included in this dataset to improve model accuracy while maintaining clinical relevance.

Building these findings, this study seeks to evaluate predictive models for diabetes readmission using a rich set of clinical, demographic, and engineered features. By examining the relative importance of various predictors and comparing model performance across algorithms, the analysis aims to contribute to ongoing efforts to improve risk prediction and inform targeted intervention strategies for diabetic patients in U.S. hospitals.

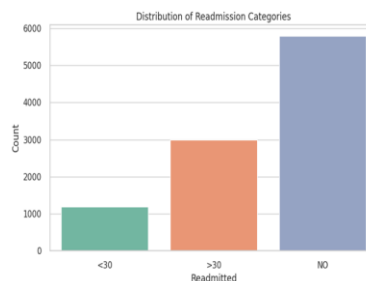


Figure 1: Most patients were not readmitted (NO), fewer returned after 30 days (>30), and the smallest group was readmitted in under 30 days (<30). This shows the data is unbalanced.

To explore the factors driving these readmission patterns, the following methodology outlines the steps taken to prepare, analyze, and model the data.

Methodology - Data Understanding

The Tableau_diabetes_data.csv dataset used in this project represents more than 100,000 inpatient encounters for diabetic patients treated across 130 hospitals in the United States over a ten-year period. The dataset contains demographic information such as race, gender, and age group; encounter-level variables including admission type, discharge disposition, and time spent in the hospital; and extensive clinical features. In addition to raw categorical attributes, the dataset also includes numerous one-hot encoded medical specialties, diagnosis groupings, and medication categories, resulting in a high-dimensional feature space. Preliminary exploratory analysis revealed severe target imbalance, with only a small proportion of patients readmitted within 30 days. Several variables contained substantial missingness, particularly weight, max glucose serum, and A1C results which limited their utility. Diagnostic fields displayed extremely high cardinality, with over 900 unique ICD-9 codes across the three diagnosis columns. These issues highlighted the need for systematic preprocessing, category consolidation, and thoughtful modeling choices.

Methodology - Data Preprocessing

Data preprocessing followed a structured pipeline and included experimentation. Because many categorical features used inconsistent placeholders such as “?”, these were standardized and mapped to an “Other” category to ensure coherence. Features with more than 40% missingness, most notably weight, payer code, max glucose serum, and A1C results were removed from the dataset, as their sparsity offered little predictive value. To improve interpretability, the age group bins were converted to numeric midpoint values. Outlier analysis revealed unrealistic values in variables such as time in hospital, number of procedures, and medication counts; these were either dropped or winsorized at the 95th percentile to preserve realistic ranges. A major challenge involved the extremely high cardinality of diagnosis and medical specialty fields. To address this, raw ICD-9 codes were consolidated into diagnosis categories, and over 70 medical specialties were regrouped into roughly nine broader groups. This step substantially reduced noise and generally made downstream models more stable. Finally, we engineered interaction features capturing relationships that clinicians often consider jointly for example, the interaction between time in hospital and number of medications, between number of lab tests and medication count, and between age and severity-related features. These interactions improved the model’s ability to capture patterns in encounter complexity and patient acuity.

Model	Recall (Class 1)	F1-Score	ROC-AUC
Random Forest	0.58	0.62	0.725
Gradient Boosting	0.59	0.62	0.723
LightGBM	0.64	0.62	0.698

Random Forest (RUS)	0.61	0.61	0.685
LightGBM (RUS)	0.64	0.61	0.694
Grad. Boost (RUS)	0.57	0.60	0.690

Figure 2. Performance Comparison of Machine Learning Models for 30-Day Diabetes Readmission Prediction.

Key Results and Insights

Model evaluation demonstrated meaningful differences in how well machine learning algorithms identified patients at risk of 30-day readmission. Figure 2 summarizes the performance of the primary models tested. LightGBM achieved the strongest recall of 0.64, outperforming Random Forest and Gradient Boosting, which achieved recall values of 0.58 and 0.59 respectively. Although LightGBM’s ROC-AUC (0.698) was slightly lower than that of Random Forest (0.725), its superior recall made it the most clinically appropriate option, as minimizing false negatives is essential when predicting high-risk patients. The F1-scores across models were relatively similar, but LightGBM offered the best balance between recall and precision among models that prioritized accurate minority-class detection.

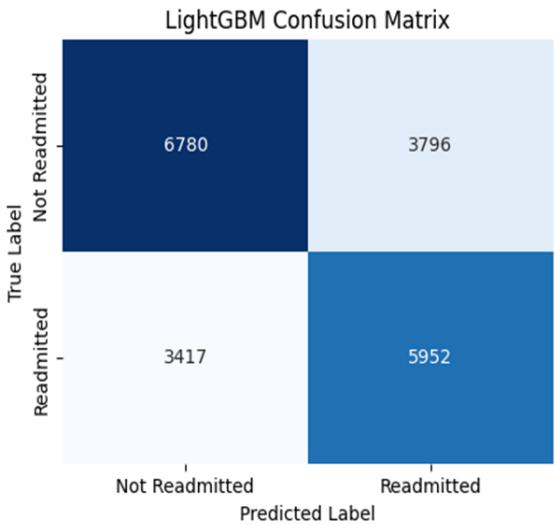


Figure 3. Confusion Matrix for the LightGBM Model Predicting 30-Day Readmission.

This confusion matrix illustrates the classification outcomes for the final LightGBM model. The model correctly identified 5,952 readmitted patients (true positives) and 6,780 non-readmitted patients (true negatives).

Further evidence of LightGBM's effectiveness is shown in Figure 3, which displays the confusion matrix for the tuned model. LightGBM correctly identified **5,952** readmitted patients (true positives) while misclassifying **3,417** (false negatives). Although this reflects the inherent difficulty of the imbalanced dataset, the model still captured the majority of true readmissions. Not-readmitted cases were also well classified, with **6,780** true negatives. These patterns confirm that the model successfully identified meaningful clinical signals even in the presence of noise, sparse features, and heterogeneous patient encounters.

Feature importance analysis indicated that variables representing encounter severity including number of labs, number of medications, time in hospital, and diagnosis category groupings were among the strongest predictors. Interaction features further improved model stability by capturing nonlinear relationships across key dimensions of patient complexity. These results align with findings in prior literature, which consistently highlight the role of encounter intensity and comorbidities in driving early readmission risk.

Conclusion & Capstone II Plan

Capstone 1 produced a fully processed and well-understood dataset, a refined feature space, and a strong predictive model for identifying diabetic patients at risk of 30-day readmission. Using a combination of category consolidation, imputation strategies, interaction features, and class-imbalance handling, we were able to train a LightGBM model. The model comparison table (Figure 2) clearly demonstrates LightGBM's received 0.64 recall, 0.62 F1-score, and superior ability to capture minority-class cases compared to other ensemble methods. The confusion matrix (Figure 3) further confirms that LightGBM identifies a substantial proportion of readmissions, making it the best-suited model for clinical prioritization. These results are consistent with findings in the literature, where LightGBM and other gradient boosting models typically achieve recall values between 0.50 and 0.65 on similar readmission datasets (Dinh et al., 2019).

Moving into Capstone 2, our focus will shift toward refining model interpretability, applying SHAP explainability techniques, expanding interaction features, and assessing fairness across demographic groups. Additional modeling strategies such as CatBoost, updated boosting pipelines, and patient-level longitudinal modeling will be explored to improve generalizability. Ultimately, the goal is to produce a clinically interpretable and operationally valuable readmission risk tool that can support better patient outcomes and more equitable diabetes care.

Together, these efforts aim to produce a clinically interpretable, operationally, useful, and equitable readmission-risk tool capable of supporting better patient outcomes and improving diabetes care delivery.

Citations

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